

STEVEN MAN

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EDUCATION

Carnegie Mellon University

Ph.D. in Robotics (QPA: 4.08/4.0)

Aug 2023 – Present

Pittsburgh, PA

- **Advisors:** Dr. Sarah Bergbreiter and Dr. Aaron Johnson
- **Fellowships:** National Science Foundation Graduate Research Fellow (2025), Tau Beta Pi Fellow (2024-2025)
- **Coursework:** Robot Dynamics Analysis, Optimal Control & RL, Robot Learning, Embodied AI Safety

University of California Santa Barbara

B.S. in Mechanical Engineering (GPA: 3.97/4.0, High Honors)

Aug 2019 – June 2023

Santa Barbara, CA

ACADEMIC RESEARCH EXPERIENCE

Carnegie Mellon University, Robotics Institute

Graduate Researcher

Aug 2023 – Present

Pittsburgh, PA

- Simulating and prototyping walking robots to study control and scaling laws of dynamic bipeds.
- Building physics-informed world-models for data-efficient model-learning.
- Applying learned safety filters to high-dimensional control systems using latent-space dynamics.
- Mentoring 6 undergraduate researchers on building various scaled biped robots and developing control strategies for stable, efficient walking over uneven terrains.

Carnegie Mellon University, Robotics Institute Summer Scholars

Undergraduate Researcher

June 2022 – Aug 2022

Pittsburgh, PA

- Created the first micro-scale Delta robot using Two-Photon Polymerization to study 3D-printing applications in micro-robots, achieving less than 5 micron trajectory tracking precision and accuracy error.
- Modeled micro-robot motion using Simscape Multibody and analyzed its trajectory using TEMA software.
- First-authored paper currently being submitted to *Science Robotics*.

Hawkes Lab, UCSB

Undergraduate Researcher

Jan 2021 – June 2023

Santa Barbara, CA

- Built robots to mimic embryo cells to study how the robot cluster can collectively change shapes.
- Designed Printed Circuit Boards and 3D-Printed assemblies to rapidly prototype embryo-inspired robots.
- Developed Arduino algorithms for light sensing and motor actuation using a photo-diode network.

Pennathur Lab, UCSB

Undergraduate Researcher

Jan 2022 – June 2023

Santa Barbara, CA

- Characterized a Liquid-Metal Microfluidic Energy Transducer to maximize electrical energy harvesting.
- Fabricated microfluidic devices using Deep Reactive Ion Etcher, Electron-Beam Metal Deposition, Atomic Layer Deposition, and Photolithography at UCSB's Nanofabrication facility.

INDUSTRY EXPERIENCE

SpaceX

Space Lasers Mechanical Engineering Intern

June 2023 – Aug 2023

Redmond, WA

- Designed, assembled, and tested a robotic optical alignment production stand to manufacture flight-level satellite hardware in 8 weeks.
- Implemented and optimized a robot automation procedure to reduce takt time by 50% compared to previous generation stands while achieving micron-precision alignment.

Johns Hopkins University Applied Physics Laboratory

Mechanical Engineering Intern

June 2021 – Sept 2021

Laurel, MD

- Built a mechatronic assembly for the NASA Dragonfly Mission Thermal Trim Device Tests to study the system's capacity to exchange heat with Titan's cryogenic atmosphere.
- Performed thermal balance tests on the device in a thermal chamber and analyzed its performance.

PROJECTS

Apical Robotics Capstone | UCSB

Sept 2022 – June 2023

- Developed the largest soft robot ever (100 ft long) for industrial pipe inspection applications and performed an onsite demonstration with the team for Bechtel Corp.
- Led the design and testing of a blower fan module capable of high-volume (60 cfm) rapid flow reversal.
- Designed and built an electronics and communication system to supply the power required for pressurized robot locomotion, transmit live video feed, and perform feedback control on pressure and growth speed.
- Winner of UCSB Mechanical Engineering Capstone Innovation Award and New Venture Competition Tech Push.

Vine Endoscopy Robot | UCSB

March 2022 – June 2022

- Designed a soft vine robot for medical endoscopic applications utilizing highly-precise fabrication techniques.
- Constructed a compact RF motor control circuit to allow for both extension and retraction of robot tip.
- Rapidly prototyped vine robot tip mounts using 3D-printing and reduced growing friction by 50%.

Ionic Skies Capstone | UCSB

June 2020 – June 2021

- Designed a gravity-driven launcher with a pulley mechanism to launch an ionic wind aircraft.
- Used Python to numerically model the relationship between take-off velocity and launch mass.

PUBLICATIONS & PRESENTATIONS

K. Nakamura, A. Bishop, **S. Man**, A. M. Johnson, Z. Manchester, A. Bajcsy. "How to Train Your Latent Control Barrier Function: Smooth Safety Filtering Under Hard-to-Model Constraints," *Learning for Dynamics & Control Conference*, 2026. (Submitted)

S. Man*, S. Kim*, S. Bergbreiter. "The microDelta: downscaling microscale Delta robot enables ultra-fast and high precision movement," *Science Robotics*, 2025.

S. Man, S. Narita, J. Macera, N. Oke, A. M. Johnson, and S. Bergbreiter, "Zippy: The smallest power-autonomous bipedal robot," *IEEE International Conference on Robotics and Automation*, 2025.

W. E. Heap, **S. Man**, V. Bassari, S. Nguyen, E. B. Yao, N. A. Tripathi, N. D. Naclerio, and E. W. Hawkes, "Large-scale vine robots for industrial inspection," *IEEE Robotics & Automation Magazine*, 2024.

S. Man, S. Bergbreiter, and A. M. Johnson. "Hip-Leg Coordinated Quasi-Passive Biped Walker for Energy Efficiency and Step Ascension," *Dynamic Walking Conference*, 2024 (Poster Presentation)

W. Heap, **S. Man**. "Novel Miniaturized Apical Extension Robot For Small Bowel Endoscopy," *UCSB ME153 Mechanical Engineering Design Fair*, 2022 (Poster Presentation)

S. Man, M. Devlin, E. Hawkes. "Embryo Cell Robot Dynamics," *UCSB URCA Poster Colloquium*, 2022 (Poster Presentation)

GRANTS & AWARDS

National Science Foundation Graduate Research Fellowship (2025)

Tau Beta Pi Fellowship (2024)

UCSB Mechanical Engineering Capstone Innovation Award (\$1500) (2023)

UCSB New Venture Competition Tech Push 3rd Place + People's Choice (\$4500) (2023)

UCSB Edison STEM Research Scholarship (\$2000) (2022)

UCSB Gene & Susan Lucas Research Fund (\$1000) (2021, 2022)

UCSB URCA Research Grant (\$750) (2021)

LA Hacks 2021, 1st Place (\$500 Hardware) (2021)

SB Hacks VII, 2nd Place (\$300 Hardware) (2021)